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PROJECTS

RAG Retrieval System for Operational Logbooks | Python, RAG, sentence-transformers, OpenAI API
Lawrence Berkeley National Laboratory dec 2025 - present

- Built end-to-end RAG system enabling natural-language search across large-scale operational logbooks using hybrid retrieval and semantic ranking
- Designed retrieval pipeline combining metadata filtering, keyword matching, sentence-transformer embeddings, and top-k retrieval workflows
- Generated grounded LLM summaries constrained to retrieved logs, improving contextual relevance over keyword-only search workflows

Boston Rental Price Dynamics | Python, scikit-learn, Random Forest, pandas

- Built ML models analyzing pricing dynamics across Airbnb listings and long-term rental markets in Greater Boston
- Engineered preprocessing pipeline with feature selection, one-hot encoding, log transformation, and imputation for skewed marketplace data
- Improved predictive performance from R^2 0.64 to 0.71 using Random Forests and analyzed key geographic/listing-level pricing signals

EXPERIENCE

Winner of Hampton's Road 2025 Datathon - Norfolk, VA
Old Dominion University - October 2025

- Lead the translation of complex ecological and modeling outputs into a clear, accessible narrative for a non-technical public-health judging panel.
- Collaborated on deep learning pipeline using probabilistic data structures and spatial signals to predict mosquito species presence.

ABM Cluster Narrative Research - Remote
Virginia Modeling Simulation and Analysis center
March -September 2025

- Built Python pipeline for NetLogo agent-based simulations, exporting tick-level agent data for clustering and behavioral analysis
- Developed K-Means + PCA clustering workflows and visualization tooling to identify representative behavioral patterns across simulations
- Designed LLM-assisted storytelling pipeline using the OpenAI API to translate simulation outputs into policy-oriented narratives for non-technical stakeholders

Marketing Director - Shreveport Louisiana
Martin Speciality Coatings
2018-2021

- Directed regional marketing campaigns leveraging analytics and targeted content strategy to expand client engagement and increase brand visibility across digital and print channels.

CORE COMPETENCIES

Python | SQL | Java | Git | Jupyter Notebook | scikit-learn | pandas | NumPy | matplotlib | seaborn | OpenAI API | sentence-transformers | semantic retrieval | embedding-based search | clustering | feature engineering | Tableau | OpenRefine | DataWrapper | NetLogo | PyNetLogo | HTML/CSS | JavaScript | Agile/Scrum | Adobe Creative Suite | French (Fluent)

EDUCATION

Old Dominion University | B.S. Data Science, May 2026 | M.S. Data Science, Expected May 2027
Western Washington University | Business Administration, Marketing Concentration, 2016